

PURDUE UNIVERSITY, IN, USA

WMO: 724386

Lat: 40.412N

Lon: 86.937W

Elev: 599

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14835

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	4.9	-8.4	3.6	0.5	-3.2	4.7	5.9	26.1	25.0	23.8	25.6	7.0	270	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	91.3	75.4	89.2	74.3	86.6	72.8	78.5	87.6	76.9	85.4	75.3	83.3	9.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	137.9	84.3	74.2	130.7	82.1	72.8	124.3	80.1	42.5	87.8	40.6	85.5	39.1	83.6	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.5	17.7	-7.8	95.3	7.7	3.3	-13.4	97.7	-18.0	99.7	-22.3	101.5	-27.9	103.9		
			-8.5	81.5	7.5	2.2	-13.9	83.1	-18.3	84.4	-22.5	85.7	-28.0	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	26.6	30.7	40.8	52.5	62.6	71.6	74.4	73.2	66.9	55.0	42.4	32.0
	DBStd	19.20	12.71	11.72	11.26	9.39	8.50	6.31	5.53	5.15	7.30	9.12	10.04	10.85
	HDD50	2577	729	547	332	78	7	0	0	0	1	50	269	564
	HDD65	5561	1190	961	753	387	151	17	2	4	64	332	678	1022
	CDD50	3496	4	6	47	154	396	649	756	720	509	206	42	7
	CDD65	1004	0	0	3	12	76	216	293	259	122	23	0	0
	CDH74	9037	0	0	23	117	709	1935	2697	2217	1133	204	2	0
	CDH80	3075	0	0	3	14	197	680	997	751	388	45	0	0
(14) Wind	WSAvg	7.7	9.3	9.3	9.4	9.7	7.7	6.4	5.6	5.3	5.8	7.3	8.2	8.4
(15) Precipitation	PrecAvg	35.6	1.7	1.5	2.6	3.7	3.7	4.1	3.4	4.1	3.0	2.4	2.8	2.9
	PrecMax	43.0	4.4	3.9	5.7	8.7	6.5	8.2	7.1	12.2	8.8	5.0	9.0	7.8
	PrecMin	29.6	0.1	0.3	0.4	0.6	1.0	0.3	0.6	0.9	0.0	0.2	0.3	0.3
	PrecStd	4.2	1.3	0.9	1.3	1.9	1.5	2.1	1.8	2.8	2.2	1.2	1.8	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	65.0	77.6	82.0	89.2	93.1	96.5	93.1	92.0	85.0	72.4	63.4
		MCWB	57.1	55.4	62.3	65.1	70.5	75.5	79.3	76.3	72.1	68.3	59.1	58.6
	2%	DB	55.3	58.8	70.6	77.5	85.1	90.1	91.3	90.2	88.2	79.9	66.4	57.4
		MCWB	52.1	53.1	58.8	62.8	68.4	73.9	77.0	75.9	71.5	65.5	58.1	53.2
	5%	DB	50.0	53.2	64.8	73.3	81.7	87.3	88.4	87.4	84.5	74.9	62.8	52.5
		MCWB	46.7	48.0	55.5	60.1	67.3	72.6	75.6	74.1	69.3	63.3	56.3	48.9
	10%	DB	43.4	46.9	59.3	69.3	77.9	84.0	85.6	84.1	81.0	70.6	58.5	46.6
		MCWB	39.9	42.3	51.3	58.1	65.6	70.9	73.7	72.3	67.5	61.3	52.2	43.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	59.3	63.9	68.8	74.4	78.5	81.8	80.1	76.1	71.7	63.5	59.5
		MCDB	60.5	63.2	75.7	78.4	84.1	89.0	91.4	89.6	87.4	80.7	68.5	63.1
	2%	WB	53.3	53.8	60.3	65.1	71.6	76.7	79.4	77.8	74.1	68.7	60.1	54.1
		MCDB	55.2	57.5	68.3	73.6	81.2	86.3	88.6	87.1	83.5	75.4	64.6	56.8
	5%	WB	46.8	48.0	57.0	62.2	69.5	75.0	77.6	76.2	72.0	65.7	57.2	48.8
		MCDB	49.1	53.0	63.8	70.6	77.7	83.8	85.3	84.2	79.8	72.2	61.7	51.7
	10%	WB	40.2	42.6	52.4	59.5	67.6	73.1	75.7	74.5	70.1	62.4	52.7	43.7
		MCDB	43.2	47.2	58.5	67.7	75.1	81.1	83.3	81.3	77.3	69.3	57.7	46.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	16.0	18.2	20.8	20.5	19.7	19.4	19.7	22.1	20.3	16.8	13.8
		MCDBR	18.7	23.1	24.5	25.2	24.3	22.9	21.3	21.8	25.6	25.4	21.4	18.7
		MCWBR	15.9	18.2	15.6	14.5	12.0	10.3	9.5	9.8	10.8	13.3	15.1	15.9
	5% WB	MCDBR	17.7	21.7	21.3	21.5	20.3	19.4	18.9	18.8	21.0	20.7	19.0	17.7
		MCWBR	16.5	18.7	14.5	14.4	11.5	9.9	9.9	9.7	10.3	12.5	15.2	16.1
(40) Clear-Sky Solar Irradiance	taub		0.299	0.326	0.350	0.404	0.434	0.441	0.459	0.458	0.422	0.364	0.321	0.299
	taud		2.427	2.339	2.359	2.218	2.188	2.208	2.157	2.147	2.227	2.384	2.454	2.451
	Ebn at Noon		274	281	286	275	269	266	260	257	258	263	263	265
	Edh at Noon		25	32	35	43	46	45	47	46	40	30	25	23
(44) All-Sky Solar Radiation	RadAvg		549	826	1155	1495	1708	1955	1939	1733	1437	981	648	456
	RadStd		58	82	89	126	153	119	107	99	110	92	67	52

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page